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The global AI race and defense's new frontier





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EXECUTIVE SUMMARY

Artificial intelligence (AI) has become a defining pillar of modern military advancements that will revolutionize strategic decision-making, surveillance, autonomous systems, and logistics. The unprecedented capabilities AI offers represent a seismic shift in how nations approach defense. Future battlefields will be shaped by AI-driven real-time data processing, autonomous decision-making, and multi-domain coordination, establishing these as distinctive features of military superiority. Currently, AI is mainly deployed in smaller, clearly scoped areas with a dedicated focus on low-hanging fruits. Larger, more complex and integrated systems are not yet broadly deployed.

Looking ahead, the most transformative impacts of AI are anticipated in three key areas in the short term: predictive decision-making, enhancing the capabilities of autonomous systems, and dynamic resource management. Within these domains, AI promises to revolutionize situational awareness, as well as operational effectiveness and readiness, becoming a critical force multiplier. Furthermore, AI is expected to become the first line of national cyber defense.

Early adopters, such as the United States and China, who are successfully integrating AI into their defense strategies, are already gaining a significant technological and organizational edge and are expected to dominate the future AI in defense landscape. In contrast, Germany and other European nations risk falling behind due to fragmented strategies, regulatory hesitation, and challenges in technological and organizational implementation.

As AI will revolutionize the defense field, it is essential for the Western alliance to comprehensively address the core ethical and legal implications, as well as the inherent target conflicts and risks, that are associated with its development. Only by ensuring that AI is responsibly deployed in the defense field, fully in line with central democratic principles and respect for human rights, can global legitimacy be ensured and trust developed. Ethical safeguards and “human in the loop” requirements are a key part of ensuring that AI applications are trustworthy and accountable. Actively shaping the future of ethical AI use to ensure deterrence and warfighting readiness is the central task in this critical phase.

Germany has the potential to take a leading role in this collaborative development of responsible AI in the defense field, building on its strong reputation for ethical governance and multilateralism. To fill this role, Germany needs to develop a comprehensive and unified national AI strategy that aligns with its industrial and civilian AI ecosystems, in close collaboration with European and NATO partners. The success of this strategy will depend on resolving data and infrastructure bottlenecks, investing in R&D and high-performance infrastructure, and addressing the skills and workforce gap. Procurement processes need to be refined, and cultural resistance tackled.

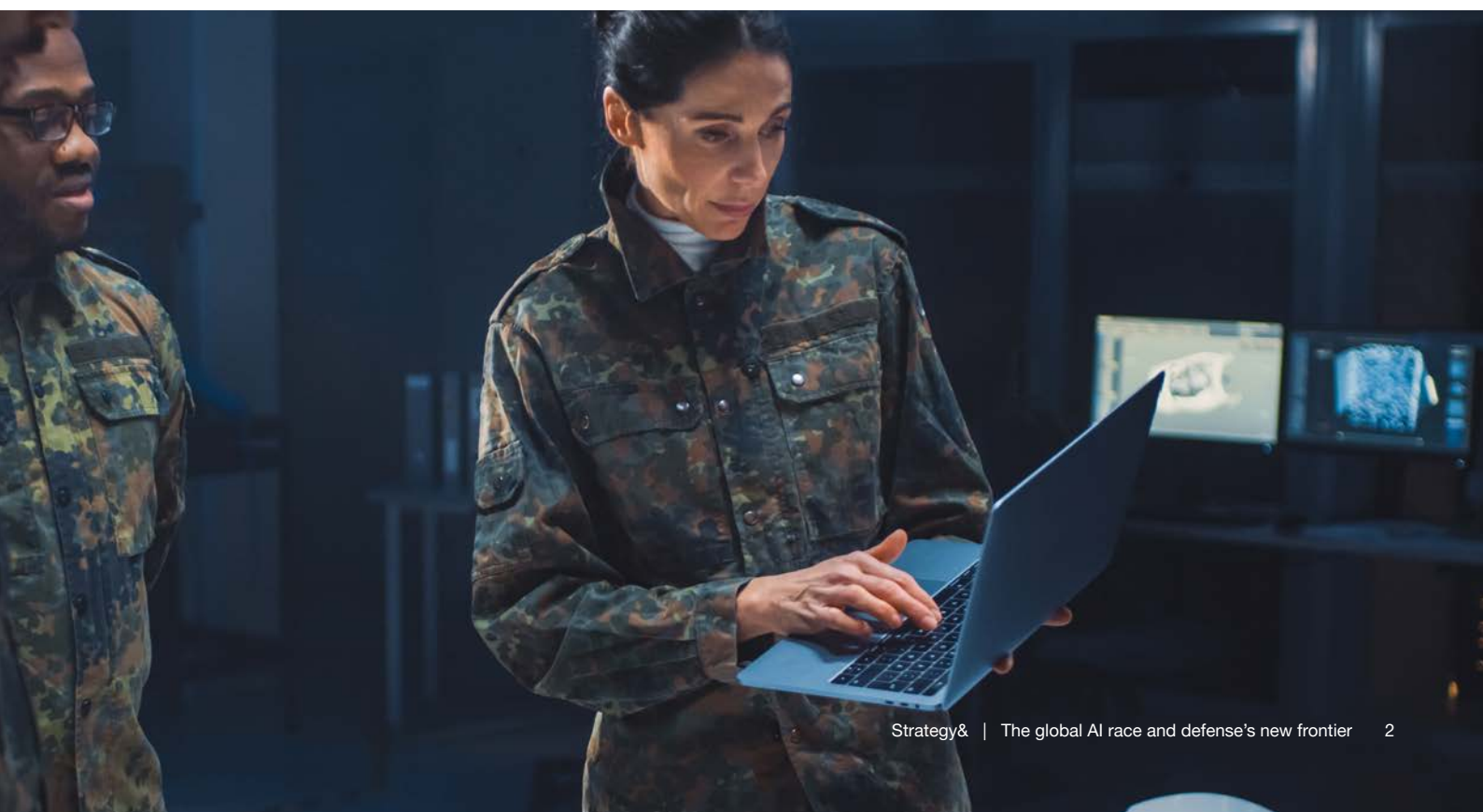
INTRODUCTION

Artificial intelligence is evolving and making a difference across the world in every industry. Its impact on defense is no less transformative. Unlike previous technological revolutions, AI is not just a tool, but an enabler of entirely new paradigms. Its applications extend beyond operational efficiency, offering capabilities that fundamentally alter mission effectiveness, speed, precision, and the scale of military operations.

This report explores the transformative potential of AI in defense, examining its impact on military capabilities and evaluating the emerging race for AI leadership. It highlights the multifaceted applications of AI, from predictive analytics and autonomous systems to cyber defense and intelligence-gathering, demonstrating how this technology is expected to be central to maintaining military superiority in an increasingly complex and interconnected global environment.

It also focuses on the essential ethical and operational challenges that accompany AI development and adoption, outlining the central necessity to comprehensively address these challenges and focus on responsible AI development in the defense sphere as a basis for global legitimacy and trust.

This report zooms in on Germany, outlining its potential to become a leading player in responsible AI in defense development, building on its longstanding reputation for ethical governance and multilateralism. It discusses central challenges ranging from infrastructural deficits to cultural resistance, and outlines a strategic roadmap to enable Germany to address these challenges and fulfill its potential.



CHAPTER 1

AI as an exponential driver of military capabilities

1.1 Defining AI in defense

Modern militaries operate within an environment of unprecedented complexity, where the volume of available data, the speed of technological change, and the sophistication of adversarial strategies continue to grow at an exponential rate. Traditional decision-making processes, often constrained by human cognitive limits, struggle to keep pace with the continuous influx of intelligence reports, sensor feeds, and cyber threat alerts saturating today's strategic and operational landscapes. In response to these challenges, artificial intelligence (AI) has emerged as a key enabler of next-generation defense capabilities, offering militaries the potential to identify meaningful patterns hidden within massive datasets, anticipate critical logistical demands, and detect hostilities before they materialize. Furthermore, multi-domain operations—integrating land, air, maritime, cyber, and space capabilities—are increasingly reliant on AI to ensure coordinated action across these interconnected arenas. AI-driven solutions promise to enhance the agility and resilience of armed forces as they contend with complex, multi-domain threats.

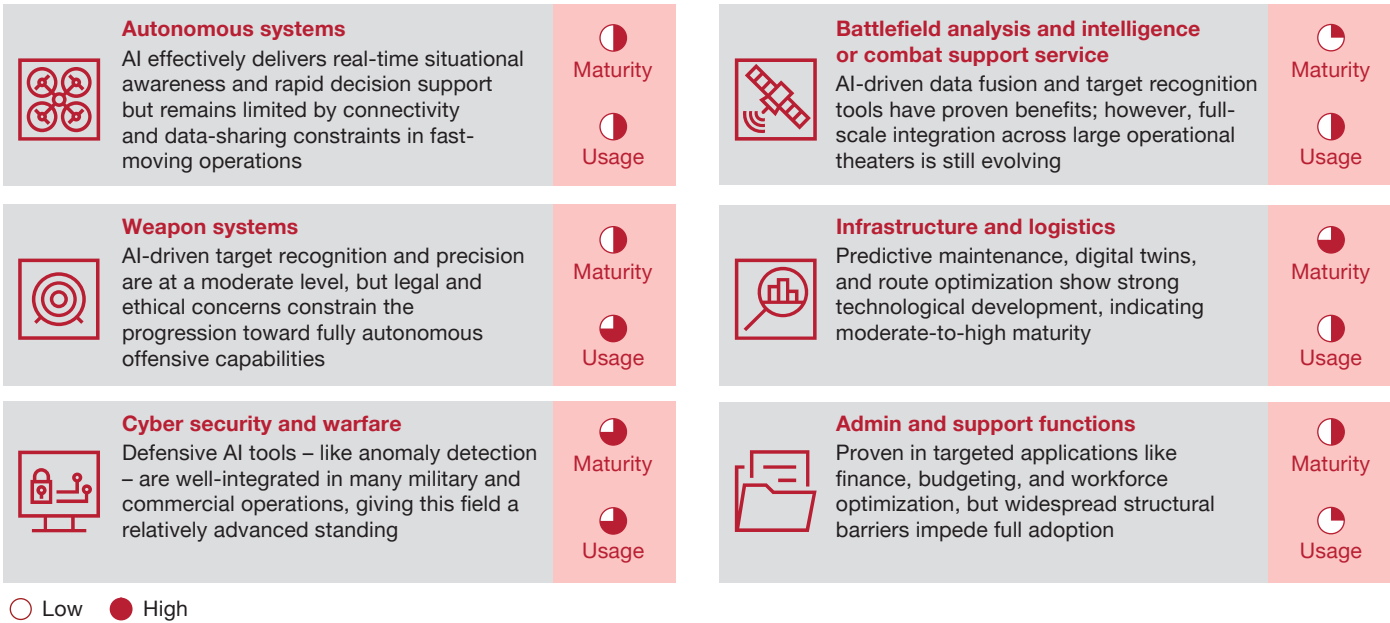
In essence, AI in the military context involves deploying algorithms, machine learning techniques, and advanced data processing frameworks to solve intricate military problems. These problems have arisen due to the enormous amount of data generated from satellites, drones, sensor networks, and other sources, inhibiting humans in sifting through and interpreting the information quickly enough. Decision support and command and control can be a bottleneck in modern warfare where AI can help militaries through predictive analytics amongst others. AI technology has enabled forces to exploit speed, scale, and predictive analytics that far surpass conventional methods of command and control.

The application of AI can be seen on land, air, sea, and even digital battlefields. Cyber operations have become particularly relevant due to rising threats in networked environments, while space assets—such as satellites—are critical for reconnaissance, communication, and navigation. AI-driven tools that safeguard and optimize these domains are now recognized as indispensable for mission success (Smith et al., 2023; Valera, 2021).^{1,2} According to NATO's guidance on emerging and disruptive technologies, these AI capabilities have already begun to alter the fundamental nature of combat operations, strategic planning, and defense logistics (NATO, 2020).³ As military forces worldwide invest in AI-driven initiatives, it has become clear that adopting this technology is no longer optional, but a necessity to maintain a competitive edge. It is apparent that early adopters might dominate AI-driven defense, creating a global divide. The cyber domain, in particular, demands heightened attention. As reliance on networked systems grows, critical military infrastructure faces escalating threats from sophisticated cyberattacks, espionage, and sabotage. AI plays a vital role in this domain by enabling advanced threat detection, real-time response mechanisms, and proactive defense strategies to counteract adversarial actions in cyberspace.

Equally significant is the role of space in modern military operations. Space-based assets, such as satellites, underpin essential functions including surveillance, secure communication, precision navigation, and early warning systems. The vulnerabilities of these assets, compounded by increasing adversarial competition in space, underscore the need for AI-driven solutions to enhance their resilience and optimize their functionality. From autonomous anomaly detection to dynamic resource allocation in space-based networks, AI is crucial for ensuring operational superiority in this high-stakes domain.

As highlighted by NATO and other defense organizations, the integration of AI into land, air, space and cyber operations represents a transformative shift that amplifies the scope and efficacy of military capabilities across all domains. Failure to integrate risks undermining the full potential of AI in defense, leaving forces vulnerable in environments where dominance is increasingly dictated by technological superiority.

EXHIBIT 1
Clustured overview of Military AI capabilities



Source: Strategy& analysis

1.2 Addressing core defense challenges through AI

Militaries today confront a host of interrelated challenges that undermine their ability to prevail in an era defined by rapid technological shifts and globalized security threats. One core difficulty stems from the increasing complexity and volume of data to be processed under immense time pressure. Human analysts, while highly skilled, cannot keep pace unaided with the sheer amount of information derived from satellites, sensors, and reconnaissance platforms. Simultaneously, adversaries employ asymmetric tactics, hybrid warfare strategies, and cyber infiltration, further straining the capacity of defense organizations to track, identify, and neutralize threats before they escalate. United States Department of Defense reports have highlighted how data overload, coupled with human decision-making latency, creates vulnerabilities that undermine operational effectiveness (U.S. DoD, 2018).⁴ The volume of data also needs to be processed in a meaningful and efficient manner, and as such the success of AI in defense relies on timely resolution of data and infrastructure bottlenecks (see *Exhibit 1, previous page*).

In response to these complex conditions, AI provides a range of interconnected solutions that bolster military performance. On the ground, advanced machine learning algorithms enable improved situational awareness and automated reconnaissance, while AI-driven target recognition systems enhance precision and coordination for aerial platforms. At sea, AI supports sophisticated detection and tracking of submarine and surface threats, thereby strengthening maritime domain awareness (UNIDIR, 2024).⁵ In space, autonomous satellite systems leverage real-time analytics to protect valuable orbital assets, and in the cyber realm, AI accelerates threat hunting, intrusion detection, and response measures (NATO CCDCOE, 2021).⁶ With such broad-ranging applicability, AI can conceivably enhance nearly every facet of modern military operations with its multi-domain potential. That said, the following sections concentrate on the application areas that exhibit significant leaps in development or display greater operational maturity, promising immediate and tangible benefits to defense stakeholders (European Defence Agency, 2021).⁷ Ultimately, AI-enhanced collaboration across military systems will determine battlefield superiority.

Autonomous systems



Autonomous systems have quickly progressed from research prototypes to operational assets capable of undertaking high-risk missions with minimal human oversight. Unmanned aerial vehicles (UAVs) like the MQ-9 Reaper are already active in surveillance and targeted strike roles, while unmanned ground vehicles (UGVs) assist in reconnaissance or cargo transport through hazardous zones. AI will be most powerful when used in autonomous systems enhancing already-existing capabilities, e.g., via autonomous location of targets. Emerging trends include “swarm” tactics, where multiple autonomous platforms collaborate to overwhelm adversaries or conduct large-scale monitoring (Nathan et al, 2023).⁸ Yet, major questions persist about the ethical and legal implications of lethal autonomous weapons systems (LAWS), particularly regarding compliance with international humanitarian law and preserving meaningful human control over the use of force (International Committee of the Red Cross, 2021).⁹ As a result, the maturity of autonomous systems spans a broad spectrum: fundamental capabilities (e.g., navigation, obstacle avoidance) are well-developed, but fully autonomous swarms and lethal systems remain at relatively early stages due to both technological complexity and legal uncertainties (Stanford University, 2022).¹⁰

Weapon systems



Weapon systems integrated with AI vary considerably in complexity and operational scope, ranging from enhanced targeting software to prototypes of autonomous offensive systems. Deep learning algorithms that refine target recognition have shown promise in reducing friendly fire incidents and civilian casualties by improving the accuracy of situational identification (United Nations, 2023).¹¹ Precision-guided munitions similarly benefit from machine learning improvements in trajectory calculation, decreasing the margin of error and thus collateral damage. However, the prospect of lethal autonomous weapons—that is, weapons capable of selecting and engaging targets with minimal human intervention—remains controversial. While AI-driven precision and targeting features display mid-level maturity, fully autonomous offensive capabilities remain constrained by a complex interplay of legal, ethical, and political factors (CCW, 2023).¹² Thus, in order to realize AI gains and to use AI's full potential, the adverse effects of fast technological advancements and time-intensive regulatory processes must be overcome.

Cyber security and cyber warfare



In the cyber domain, AI has emerged as both an essential shield and an evolving sword (Carnegie Endowment for International Peace, 2022).¹³ Defensive measures increasingly rely on machine learning to identify malicious network traffic, adapt firewall rules, and coordinate rapid responses to zero-day exploits. On the offensive side, AI enhances penetration testing, infiltration techniques, and advanced persistent threat (APT) detection, although regulatory and policy frameworks are slower to adapt to the ethical complexities of AI-assisted offensive cyber operations (Politico, 2024).¹⁴ Maturity in AI-driven cyber defense tools can be considered moderate-to-high, with many commercial and military sectors already integrating solutions into their security operations centers. Today, cyber is recognized as the fifth dimension of warfare (EDA, 2025)¹⁵ and will eventually become the first line of national defense. Nonetheless, the speed at which cyber threats evolve requires ongoing research, substantial investment in specialized AI talent, and dynamic governance that can pivot as adversaries innovate new attack vectors.



Battlefield analysis and intelligence/ combat support services



AI-driven analytics have proved their value in battlefield analysis and combat support roles by streamlining data fusion, target recognition, and real-time mapping in complex operational environments (Army University Press, 2024).¹⁶ Ground-based sensors, aerial drones, and satellite imagery collectively produce massive datasets that exceed human capacities for timely processing (King, 2024).¹⁷ Through AI, militaries can automate the classification of incoming data, forecast enemy maneuvers, and generate battle damage assessments with heightened speed and accuracy (King, 2024).¹⁸ Although partial implementations have already yielded significant benefits, particularly in localized theaters, complete integration across wide areas remains challenging due to inconsistent connectivity, competing classification protocols, and interoperability shortfalls among allied forces. As a result, the maturity level for battlefield analytics is moderate, reflecting incremental but meaningful advancements that still rely on improved multi-domain data management.

Infrastructure and logistics



Infrastructure and logistics have emerged as key beneficiaries of AI-enabled innovations, particularly due to the extensive data generated by maintenance schedules, supply chains, and transportation networks, including for military manufacturing. AI has the potential to use this data to transform logistics and maintenance and become a critical force multiplier as a result. Predictive analytics and digital twin technologies now make it possible to anticipate equipment failures on critical military platforms. These insights facilitate proactive maintenance and more efficient inventory management, thereby reducing costly operational downtime. The use of AI-driven route optimization further streamlines the movement of goods and personnel, offering tactical advantages in contested environments where rapid and precise logistical support can be decisive. Supermarkets are already actively using AI-enhanced route optimization. While these applications already demonstrate moderate-to-high maturity in limited-scale deployments, the full potential of AI-driven logistics hinges on robust data-sharing protocols, consistent interoperability standards, and the sustained commitment of leadership to fund and adopt next-generation software capabilities.



Ultimately, AI-enhanced collaboration across military systems will determine battlefield superiority.”

Administrative and support functions



Although less visible than frontline tasks, administrative and support functions underpin the overall effectiveness of any defense organization. AI plays a central role in automating financial processes, auditing defense budgets, and allocating resources with greater accuracy (Microsoft, 2024).¹⁹ Machine learning models can also optimize workforce distribution by identifying skill gaps, forecasting retirement patterns, and recommending staff realignments—a capability particularly valuable in large, complex institutions (Baabdullah, 2024).²⁰ Despite clear benefits, the adoption of AI in administrative domains faces cultural and structural barriers; many organizations remain hesitant to overhaul traditional hierarchies or share sensitive data widely, especially in environments with strict security classifications. Consequently, the current maturity of AI in administrative and support roles stands at a mid-level: it has proven its utility in targeted applications but has not yet achieved the broad operational footprint that could maximize its transformative potential.

These fields provide a tangible glimpse into the application potential of AI, as key areas where AI can yield immediate operational benefits and forge pathways to sustained strategic advantage. But AI adoption, maturity, and usage in active battlefield contexts varies. Some areas—such as cyber defense and logistics—already demonstrate relatively high levels of technological development and operational deployment. Other fields, notably fully autonomous systems and AI-assisted administrative functions, remain at mid-level maturity due to ethical, regulatory, and structural barriers. Overall, while AI’s transformative potential is widely recognized, its full integration hinges on resolving data, interoperability, and organizational challenges.

1.3. Case study—AI application

The rapid adoption of AI in military operations has transformed how wars are fought, introducing innovative tools and strategies that redefine both tactical and strategic decision-making. The war in Ukraine and the conflict in Gaza demonstrate the practical applications and potential of AI in active warfare, as they highlight how AI technologies have been deployed in real-world scenarios and offering critical insights into their capabilities, impact, and evolving roles in modern combat.



Case study: AI applications in the war in Ukraine

Ukrainian forces have leveraged AI-enhanced technologies to reshape the battlefield environment, adapting swiftly as both sides incorporate more sophisticated tools. Since 2022, reporting has consistently highlighted how Ukraine's innovative use of AI-enabled drones, dynamic cyber defense strategies, and intelligent logistics management has helped offset Russia's numerical and technological advantages.

Unmanned aerial vehicles (UAVs) and unmanned ground vehicles (UGVs) integrate machine learning algorithms that can rapidly interpret video feeds, identify equipment and personnel, and differentiate between high-value targets and decoys. AI-driven swarm tactics have matured beyond initial experiments, enabling coordinated simultaneous drone strikes which overwhelm Russian air defenses (The Economist, 2024).²¹

Ukraine's efforts to leverage AI in the cyber domain have proven equally consequential. Military units, supported by allied cyber defense teams and private sector partners,

employ AI-driven anomaly detection systems to sift through vast volumes of network traffic and communications. These algorithms identify suspicious patterns and potential infiltration attempts, enabling operators to isolate, investigate, and counter cyber threats in near real time. Defensive cyber measures have matured considerably, detecting and neutralizing malware campaigns and information operations aimed at eroding public morale (BBC, 2023).²²

Ukraine's ability to sustain high-tempo operations with limited resources hinges increasingly on AI-powered logistics solutions. Predictive maintenance algorithms anticipate equipment failures, informing commanders when to rotate vehicles, aircraft, or artillery pieces out of service for rapid repairs, to minimize downtimes. Meanwhile, emerging battlefield management software integrates intelligence from drones, satellites, and ground sensors with supply chain inventories, enabling just-in-time deliveries of ammunition, spare parts, and medical supplies.

Implications of AI application in the war in Ukraine

The democratization of AI-driven warfare has become increasingly evident. Off-the-shelf technologies and commercially-available AI applications have empowered Ukrainian forces to counter an adversary with greater numerical strength and advanced conventional weaponry in cost-effective ways that were difficult to imagine before 2022.

Integration of AI continues to shape tactics, doctrines, and procurement priorities. Analysts from the Royal United Services Institute have noted that the lessons emerging from Ukraine will influence defense investments and doctrinal development well beyond this conflict, guiding how militaries worldwide incorporate AI into their future force designs. There is growing consensus that the war in Ukraine stands as a pivotal proving ground for battlefield AI.

Finally, the war's progression suggests that future conflicts will blend kinetic and non-kinetic operations more seamlessly than ever before. AI-facilitated intelligence and cyber operations will intersect with physical combat, enabling rapid decision-making and dynamic shifts in strategy.

Case study:

Combat precision and ethics in AI deployment in the conflict in Gaza

The conflict in Gaza provides a distinct perspective on how AI can reshape modern warfare, particularly in densely populated and urban combat environments. Unlike the asymmetric field innovations observed in Ukraine, Israel's approach is rooted in a centralized, highly developed AI ecosystem supported by longstanding technological investments and close civil-military collaboration to integrate AI into intelligence gathering, precision targeting, and operational planning, as algorithmic decision-making refines urban combat doctrine while seeking to minimize civilian harm (The Economist, November 2021; The Guardian, May 2021).^{23,24}

Israeli forces have employed AI-driven surveillance systems capable of sifting through vast streams of data from satellites, drones, and ground-based sensors. Machine learning algorithms identify patterns, flagging anomalies and potential threats for human analysts to review. These tools generate a comprehensive, real-time battlefield picture, and also enhance situational awareness in complex urban settings (in part via facial recognition and behavior analysis software) to reduce the time between detection and response (Foreign Policy, August 2021).²⁵

A central hallmark of Israel's AI strategy in Gaza has been the use of algorithms to inform precision strikes. AI models trained on multispectral imagery, radar signatures, and historical targeting data help distinguish militants from civilians and high-value assets from non-combatant infrastructure. Machine-speed analysis of these inputs allows commanders to ensure that kinetic actions are proportionate and discriminate, adhering to international humanitarian law and reducing collateral damage. AI-driven simulation tools also assist in operational planning, allowing Israeli planners to test the outcomes of various tactical scenarios before execution.

Israeli forces have operationally deployed autonomous platforms that integrate AI with precision-guided munitions. The Harop loitering munition can linger over an area until its AI-driven targeting system identifies a validated threat and an attack is executed, without requiring continuous human guidance. AI is also integral to the Iron Dome air defense system. Advanced algorithms calculate trajectories of incoming rockets and artillery shells in real time, prioritizing which threats to engage (those targeting populated areas or critical infrastructure) and conserving valuable interceptors while protecting civilians.

Implications of AI application in the conflict in Gaza

Israel's experience demonstrates that AI can serve as a force multiplier, enabling a highly networked and technology-savvy military to achieve mission objectives with fewer personnel and reduced collateral damage. The deliberate integration of ethical parameters within AI targeting algorithms illustrates the feasibility of embedding moral considerations into automated decision-making processes, and Israel has itself passed policies on AI use in terms of regulations and ethics (Ministry of Innovation, Science and Technology, 2023).²⁶

The enhanced efficiency and effectiveness of Israeli operations sets an example for other technologically advanced militaries seeking to refine their conduct in urban combat environments. The Israeli model highlights the importance of transparent oversight mechanisms, continuous testing, and the involvement of legal and moral authorities to prevent misuse or overreliance on automated systems. However, it remains an open question how far ethical frameworks, policies and international laws for AI are keeping up with the rapid development and use of AI in military contexts.

Prospects and challenges

As urbanization accelerates worldwide, conflicts increasingly unfold in densely populated regions, making the Israeli approach a potential blueprint for future military operations in similar settings. AI's ability to furnish real-time intelligence and guide precision strikes suggests that future conflicts may emphasize discrimination, proportionality, and operational restraint. However, success in these scenarios will depend not only on the sophistication of AI algorithms but also on the quality of the underlying data and the robustness of ethical safeguards.

As illustrated by the experiences in both Israel and Ukraine, the rise of AI in warfare heralds a profound transformation in how states conceive, plan, and execute military operations. While technology can enable greater precision, faster decision-making, and reduced manpower requirements, it also raises questions about accountability, escalation control, and the long-term implications of delegating lethal choices to machines. For militaries seeking to maintain strategic relevance in an era defined by complex, multi-domain threats, mastering AI's potential while addressing its challenges will be key.

Early adopters will dominate AI-driven defense, creating a global divide



The success of AI in defense depends on resolving data and infrastructure bottlenecks

AI-enhanced collaboration across military systems will determine battlefield superiority

AI will be most powerful when it is used in autonomous systems, enhancing already-existing capabilities

In order to realize AI gains and to use the potentials AI creates, the adverse effects of fast technological advancements and time-intensive regulatory processes must be overcome

AI-driven cyber defense will become the first line of national defense

AI will transform logistics and maintenance, becoming a critical force multiplier

CHAPTER 2

Comparative analysis

Framework for comparative analysis

Given the vast potential of AI in defense and its current battlefield applications, identifying who is competitive in the global AI defense race in today's multi-polar and multi-crisis environment is central to gaining a deeper understanding of the strategic priorities, technological advancements, and competitive dynamics shaping the future of military capabilities worldwide (see *Exhibit 2*).

- **AI-readiness:** This includes the technological maturity, more concretely the sophistication of AI technologies developed and deployed, and the Integration into Military Doctrine, i.e. the extent to which AI has been infused into defense strategies and combat operations.
- **Strategic autonomy:** The ability of independent development and deployment of AI technologies, without reliance on foreign suppliers (incl. the scale and focus of investments into AI research, particularly defense-specific applications).
- **Ethics and governance:** How to balance the drives of innovation with ethical considerations and global norms.

EXHIBIT 2 Country comparisons

Indicative	 United States	 China	 Russia	 Germany
 AI readiness	• Leading in autonomous systems, cybersecurity and battlefield analytics • AI deeply embedded into doctrine, e.g. within JADC2, as a key enabler of rapid decision making • Clear focus on fast implementation with AI RCC¹	• Rapid advancements in AI surveillance, cybersecurity, and UAVs • Deeply embedding AI into doctrine as an enabler of, amongst others, information dominance and rapid decision-making	• Strong focus on military-specific AI, with an emphasis in the cyber domain (building on existing capabilities) in addition to UAVs and decision-making • Embedding AI into military doctrine, with a clear role in "new generation warfare"	• Behind in defense AI development, with innovation often focused on civilian/ dual-use applications • Fragmented adaption: limited coordination across defense programs, further hampered by slow procurement cycles
 Strategic autonomy	• World-leading AI-eco-systems backed by enormous public and private investment • Minimal reliance on foreign suppliers	• Significant state-led programs (e.g., Next Generation AI Plan) to ensure investment • Focus on indigenous innovation to reduce dependence on foreign suppliers	• Pursues self-reliance in AI defense tech to mitigate impact of foreign restrictions • Brain drain, limited private high-tech sector, and reliance on foreign suppliers impedes strategic autonomy	• Relies heavily on NATO/ EU frameworks • Strongly limited independent AI development, mainly within civilian domain
 Ethics and governance	• Ethical guidelines and principles exist, but criticized for prioritization of strategic advantage	• Strategic goals override ethical considerations in AI surveillance and weapon development	• Ethical questions are considered secondary to strategic advantage	• Strong and overarching commitment to ethical governance, even more propound within AI sphere

○ Low ● High

1. AI Rapid Capabilities Cell
Source: Strategy& analysis

Overarchingly, the United States retain the pole position in AI defense innovation. The United States is following a clear vision, backed by enormous private and public investment and strategic autonomy. The recent announcement of the Stargate project clearly highlights this unique ecosystems and scale of investment. The mutual interdependency between government agencies, such as DARPA, and private high-tech giants like Google, Microsoft, and Palantir facilitates the rapid development and implementation of cutting-edge AI technologies in defense applications, ensuring that the United States remains at the forefront of military innovation. With the recent announcement of an “Artificial Intelligence Rapid Capabilities Cell”, aiming to execute AI pilots within the military on an accelerated timeframe, the United States DoD is also outlining a blueprint for the rapid and agile operational implementation of AI within the military.

While the lack of transparency significantly hampers to ability to outline the Chinese capabilities on a granular level, it is clear the Communist Party of China sees AI as a strategic enabler toward military parity-and eventually superiority-over the United States. The Chinese strategy for AI is focused on state-driven initiatives, fast deployment, and putting technology first above ethics. Companies like DeepSeek show that China is rapidly advancing in developing of (civilian) AI, highlighting their innovative capabilities in this field. The centralized, state-led approach underlines the efficacy of rapid decision-making and mobilization of resources, but also raises ethical and geopolitical concerns.

Russia’s approach to AI is characterized by a strategic focus on military applications, tactical adaptation to emerging technologies, and leveraging AI for geopolitical influence. The Russian government views AI as a critical component in maintaining and enhancing its military capabilities and global standing. While Western sanctions may be hampering rapid technological advancement, these are also driving a clear focus on domestic R&D and the potential of AI technologies to bridge gaps in conventional capabilities.



Among those leading countries, the German position is unique. It is defined by a clear and longstanding commitment to ethical governance and multilateralism, while being faced with significant technological, structural and cultural barriers. While the overall caution and aversion to militarization (of AI), is well rooted within the nation’s historical experience and societal values, it also presents unique challenges in navigating the rapid pace of global AI advancements. Only by developing a clear strategy of AI in defense that builds on Germany’s reputation and is fully in line with its values, Germany can take a lead position and evade the risk being strategically irrelevant in an AI-driven defense landscape. The core reasons for Germany’s lagging in AI defense innovations are embedded in several mutually reinforcing factors, including but not limited to:

- **Fragmented Efforts:** AI initiatives are scattered among different agencies and EU programs, resulting in inefficiencies and a lack of a uniform strategy.
- **Infrastructure Deficits:** Insufficient data centers, underdeveloped cloud computing, and a lack of edge computing infrastructure hamper Germany’s capability to process and deploy AI systems.
- **Cultural Resistance:** Skepticism from the public about militarization is entwined with an entrenched risk-avoiding attitude, making the pace of AI adoption in defense slow.
- **Regulatory Barriers:** It might be implied that complex procurement processes and stringent ethical guidelines-although commendable-may stifle innovation and delay deployment.

Other nations shaping AI in defense

Countries such as Israel, the United Arab Emirates (UAE), and India are using their agility, strategic investments, and expertise in niche areas to establish positions for themselves in the AI defense ecosystem:

Israel	As a leader in AI-driven cybersecurity and autonomous systems, Israel exemplifies the benefits derived from aligning defense R&D with commercial innovation. With extensive expertise in the deployment of AI in weapons systems and battlefield application, Israel is becoming a leader in the AI defense field.
UAE	Major investments in AI and partnerships with global tech firms have turned the UAE into a place for AI innovation.
India	Still developing its AI ecosystem, Indian talent and defense partnerships offer remarkable potential for improvement.



Based on the comprehensive benchmarking, key takeaways can be identified that are required for an effective AI in defense implementation:

1

At first, the application of AI in defense requires a dedicated strategic approach, that is not just fully in line with the overarching defense architecture of the country/its alliance and its core democratic norms, but that is furthermore building on its distinct (technological) ecosystem. Without a tailored strategic approach, AI in defense will not just face key questions of legitimacy but will remain fragmented.



2

Secondly, the success of AI in defense depends heavily on the baseline technological ecosystem, more explicitly on resolving data and infrastructure bottlenecks. Without the required technical fundamentals, a strategic and widespread AI application in defense is unlikely to materialize.

3

Thirdly, a holistic overview over key application barriers along the whole AI in defense ecosystem is required to ensure a successful and widespread implementation. Without comprehensively addressing central barriers, the application of AI in defense will remain sub-scale.

For Germany, central aspects stand out:

1

At first, trying to match AI advancements of China and the US, based on their strong position, will be neither possible nor instructive for Germany. Instead, Germany should build on its ethical governance reputation to lead in the responsible AI development in the defense domain by forming European partnerships.

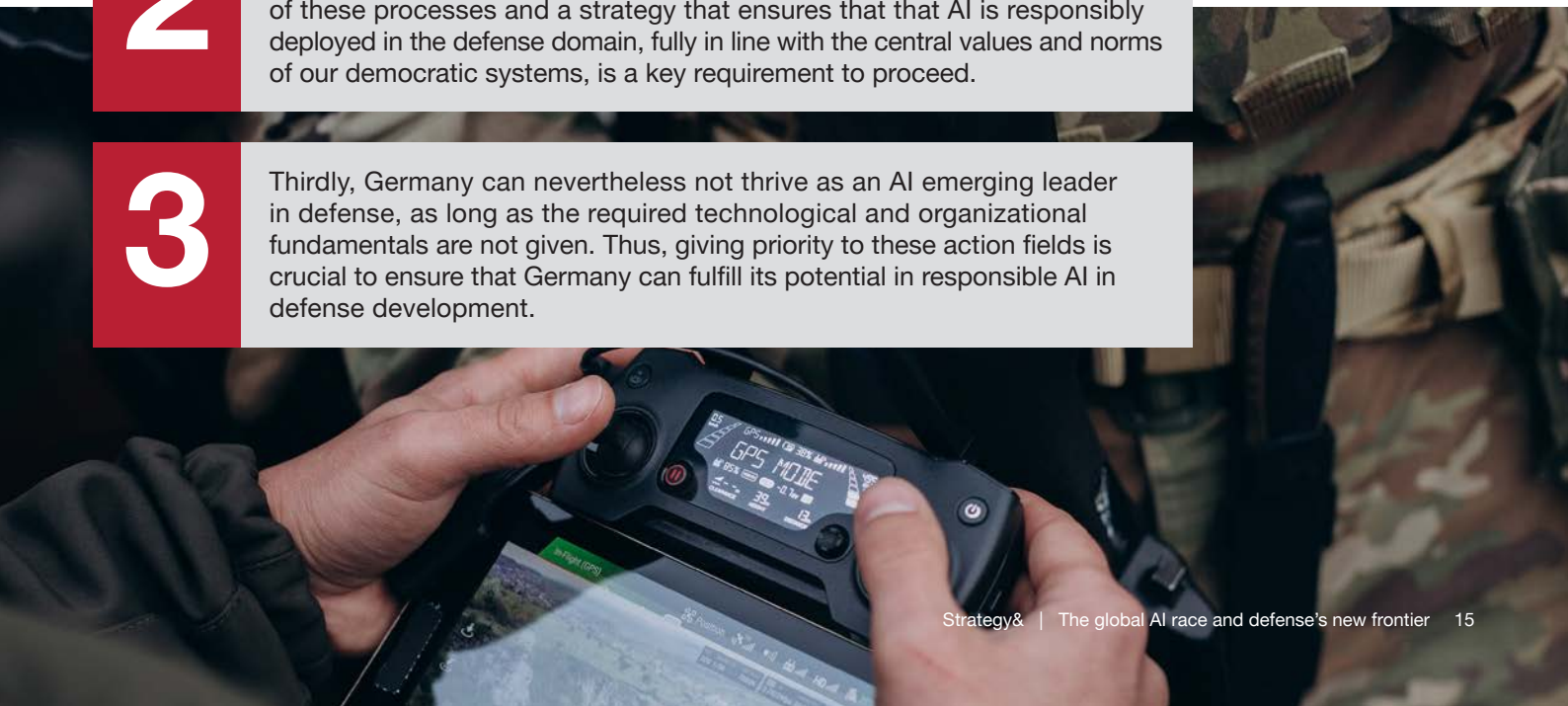


2

Secondly, procurement process and cultural resistance pose greater barriers in Germany than the pure technological fundamentals. As such, a reform of these processes and a strategy that ensures that AI is responsibly deployed in the defense domain, fully in line with the central values and norms of our democratic systems, is a key requirement to proceed.

3

Thirdly, Germany can nevertheless not thrive as an AI emerging leader in defense, as long as the required technological and organizational fundamentals are not given. Thus, giving priority to these action fields is crucial to ensure that Germany can fulfill its potential in responsible AI in defense development.



CHAPTER 3

AI scenarios for the future battlefield

In view of current AI applications and the strategic ambitions for AI, the following chapter outlines future AI scenarios for the battlefield and sets out central ethical and legal considerations, as well as application barriers and risks, associated with the proliferation of AI in the defense field.

3.1 Vision and impacts of AI-driven defense

The integration of AI into defense systems is transforming the landscape of military operations, outlining a future of enhanced efficiency, precision, and adaptability. By 2030, AI technologies are expected to play a pivotal role in reshaping how defense organizations manage resources, make decisions, and execute complex missions across domains. From optimizing supply chains and automating battlefield operations to empowering decision-makers with predictive insights, AI is poised to become an indispensable force multiplier. At the same time, the deployment of AI in defense is associated with central risks and inherent target conflicts, which increase the threat of strategic fragmentation and an associated stagnation of AI deployment in defense systems. Therefore, AI utilization needs to be carefully evaluated and deliberately managed to ensure that AI in defense is deployed fully in line with the central values of the democratic norms and systems of the Western alliance. For example, it will be crucial to set up and implement ethical safeguards whenever AI-driven decisions are made. Also, as not every country gives similar weight to AI regulation, incidents where AI systems will not be controllable or will not act as previously anticipated (or will even be misused) are likely to occur. Such events can lead to legal and compliance issues in the defense environment, and must be carefully handled. Active shaping of the future of ethical AI use to ensure deterrence and warfighting readiness is the central task to maintain global legitimacy and trust.

The following sections explore the key areas where AI's impact will be most profound, before the central risks and target conflicts to be considered are outlined.

Predictive decision-making

Fast and effective decision-making has always been a cornerstone of military success, distinguishing leaders and forces in high-stakes operations. By 2030, AI is expected to redefine this critical capability, enhancing situational awareness, accelerating decision-making processes, and improving the precision of strategic choices (Atlantic Council, 2022).²⁷ The VUCA framework—Volatility, Uncertainty, Complexity, and Ambiguity—captures the unique challenges of military environments, and AI is rapidly becoming a transformative tool for addressing these challenges.

Volatility, defined by rapid and unpredictable changes, demands real-time adaptability. AI mitigates volatility by aggregating and analyzing vast amounts of data from satellite imagery, drone feeds, sensor networks, and other intelligence streams. Unlike traditional methods reliant on human analysts, AI processes this data at unprecedented speeds, detecting patterns and alerting commanders to emerging threats. For example, during sudden battlefield changes, AI can provide actionable insights in real time, enabling timely resource deployment or strategic adjustments. AI-powered autonomous systems, such as drones, further ensure continuity by adapting dynamically to volatile environments.

Uncertainty arises from incomplete or unclear information, making it difficult to anticipate enemy actions or understand the operational landscape. AI enhances situational awareness by integrating data from diverse sources into a cohesive picture and leveraging predictive analytics to model potential outcomes (U.S. DoD, 2023).²⁸ These tools reduce uncertainty, enabling commanders to shift from reactive problem-solving to proactive strategy formulation. By identifying opportunities and threats, AI empowers decision-makers to anticipate enemy actions and select optimal responses that align with mission objectives (Atlantic Council, 2022).²⁹

Complexity stems from the interconnected nature of modern military operations, including coalition dynamics, logistics, and multi-domain strategies. AI simplifies complexity by optimizing systems, reducing cognitive burdens, and enabling cohesive coordination across land, air, sea, space, and cyber domains. Through real-time analysis, AI streamlines logistics, ensuring resources are delivered efficiently even in complex, resource-intensive environments. Its ability to simulate and analyze various scenarios allows commanders to evaluate and execute integrated strategies that address the multifaceted demands of modern warfare.

Ambiguity, caused by conflicting or unclear information, poses risks of misinterpretation and ineffective responses. AI addresses ambiguity through advanced pattern recognition, distinguishing between genuine threats and deceptive maneuvers. By testing strategies in simulated environments, AI ensures robust decision-making in the face of ambiguous data. Furthermore, its capability to provide probabilistic analyses helps decision-makers weigh options and make informed choices, even when clarity is lacking.

AI's transformative potential lies not only in its ability to address the challenges of volatility, uncertainty, complexity, and ambiguity but also in its capacity to revolutionize decision-making cycles. By streamlining and accelerating decision processes, AI ensures that battlefield strategies are executed in a timely and adaptive manner (Atlantic Council, 2022).³⁰ Ultimately, AI-powered predictive decision-making represents a leap forward in military strategy, enabling faster, more informed, and more precise operations in increasingly dynamic environments. As the defense sector continues to embrace these technologies, they will play a pivotal role in shaping the future of global security.



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Collaborative autonomous systems

The defense sector is poised to witness a paradigm shift in how autonomous systems collaborate across domains. While autonomous systems are already operational and continuously advancing, the next decade will focus on their seamless integration and coordination across multiple domains. This transformation will enable synchronized operations that leverage the strengths of various systems in a cohesive manner.

For example, swarms of drones operating in tandem with naval assets can perform complex, synchronized tasks such as reconnaissance, supply delivery, and precision strikes, enhancing mission efficiency and effectiveness. Such swarms are particularly valuable for amplifying multi-domain awareness, as their real-time data collection and analysis offer unparalleled situational understanding and broader operational coverage (DARPA, 2023).³¹

Moreover, AI-driven coordination tools will allow allied forces to synchronize their efforts, sharing real-time data across air, land, sea, and cyber domains. This multi-domain integration not only ensures the cohesive execution of joint missions, but also enhances decision-making processes under dynamic and complex conditions (NATO, 2024).³²

The increased deployment of drones and other unmanned autonomous systems offers the additional benefit of reducing the need for personnel in high-risk environments. By automating tasks that traditionally require human operators, these systems not only minimize casualties but also free up resources for strategic operations (RAND Corporation, 2019).³³

Furthermore, AI-powered autonomous systems significantly enhance the accuracy of target identification and engagement. Leveraging machine learning and advanced sensor fusion, these systems can reduce collateral damage and improve mission success rates, aligning with ethical and operational imperatives (Air&Space Forces, 2021).³⁴

In sum, collaborative autonomous systems are set to redefine modern warfare by 2030, offering enhanced operational efficiency, multi-domain synergy, and a reduced human footprint on the battlefield. These advancements underline the transformative role of AI in defense and its potential to shape a more strategic, intelligent, and resilient military future. It also raises key ethical and legal concerns that need to be carefully analyzed and built into the baseline for responsible development.



Dynamic resource management

By 2030, dynamic resource management powered by AI is set to revolutionize defense logistics and operational efficiency. AI-driven logistics systems will transform military supply chains by streamlining inventory management and transportation networks. These systems will optimize the delivery of human and material resources to where they are needed most, ensuring precise supply allocation and reducing delays that can jeopardize mission outcomes.

Dynamic resource allocation, facilitated by advanced algorithms, will maximize the efficient use of fuel, ammunition, and critical supplies, minimizing waste and enhancing sustainability. For instance, AI systems can analyze consumption patterns and operational demands in real time, ensuring that resources are allocated effectively across multiple domains (RAND Corporation, 2019).³⁵

AI tools will also revolutionize personnel scheduling, improving the efficiency of deployment and ensuring that mission-critical roles are consistently and optimally staffed. This capability is particularly vital for operations requiring rapid adaptability to changing circumstances, such as disaster relief or high-tempo combat missions (Developments and Advances in Defense and Security, 2024).³⁶

Predictive maintenance, underpinned by AI analytics, will further enhance operational readiness. By leveraging data from sensors and historical maintenance logs, these systems can identify potential equipment failures before they occur. This capability not only reduces unplanned downtimes but also extends the operational lifespan of critical assets, ensuring higher equipment availability and cost-effectiveness (DARPA, 2023).³⁷

In sum, AI-driven dynamic resource management will be a cornerstone of defense innovation, delivering unparalleled efficiency, sustainability, and resilience. These advancements underscore the transformative potential of AI to empower military operations with precision and agility in an increasingly complex global security environment.

As the defense sector evolves, AI will be at the heart of its transformation, offering unmatched capabilities in multi-domain collaboration, resource optimization, and strategic foresight. By enabling faster, more informed, and efficient operations, AI-driven technologies will enhance mission success rates while minimizing human risk and resource wastage. The innovations envisioned not only promise to redefine military strategy but also underscore the critical need for ethical, secure, and sustainable integration of AI in defense. As these advancements unfold, they will shape a more intelligent and resilient global security landscape.

3.2. Ethical, legal considerations and inherent target conflicts

The potential of AI in defense is undeniably vast, offering transformative capabilities that could redefine national security and warfare. However, it is equally clear that the adoption and application of AI in defense comes with inherent target conflicts and profound ethical and legal implications. These issues are particularly evident in the realm of AI-powered autonomous weapons. An extreme example of this is an AI system making an independent decision resulting in civilian casualties. Apart from the obvious ethical questions and implications, the determination of accountability is not yet defined. It could lie with the developers of the technology, the military personnel deploying it, or the policymakers sanctioning its use. Although this fundamental risk is noted and concern shared across the globe, competing national security interests and the race for AI leadership are currently severely hampering the development of a global regulatory framework for the use of AI, not just in military but equally in civil applications (United Nations University, 2023; Carnegie Endowment for International Peace, 2024).^{38,39}

In addition to the challenges relating to application, even the training of AI systems presents significant target conflicts, particularly concerning data security and privacy. High-quality, reliable data is indispensable for the effective training and refinement of AI applications. Yet, the data types in question are often subject to stringent regulations and restrictions, especially when they involve sensitive information critical to secrecy and security. This creates a paradox: the need for unrestricted access to robust datasets clashes with the imperative to safeguard data against breaches, misuse, or exposure. Balancing these conflicting priorities is an ongoing struggle, highlighting the broader tension between advancing technological capabilities and adhering to ethical, legal, and security standards.

Thus it is imperative for the Western alliance to take proactive measures to guarantee that the deployment of AI in the defense sector aligns with the foundational principles and values of democratic systems. The responsible use of AI in this context must reflect a commitment to transparency, accountability, and respect for human rights, ensuring that these technologies are used in ways that bolster—not undermine—global trust and legitimacy. To achieve this, shaping the future of ethical AI use is not merely a technical challenge, but a strategic necessity. Striking a balance between maintaining deterrence and warfighting capabilities while upholding democratic norms is essential to reinforce the credibility and standing of the alliance on the world stage.



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A key pillar in this endeavor is the continuous implementation of “human-in-the-loop” requirements, which ensure that human operators maintain oversight and ultimate decision-making authority over critical AI systems, particularly those with lethal capabilities. This approach is central to fostering public confidence in AI technologies and safeguarding against unintended consequences or unethical outcomes. By embedding human judgment at critical points in the decision-making process, the Western alliance can demonstrate a clear commitment to accountability and ethical governance.

Moreover, the adoption of such safeguards provides a framework for distinguishing democratic uses of AI from those employed by authoritarian regimes, which may prioritize efficiency and control over ethical considerations. In doing so, the Western alliance can set a powerful example, advocating for a global standard that emphasizes the integration of ethical principles into the development, deployment, and operation of AI systems in defense. This effort will be instrumental in establishing trust among international partners and reinforcing alliances in the face of rapid technological advancement.

3.3 Central application barriers

Apart from the central ethical and legal implications and the inherent target conflicts, key and concrete application barriers are evident that hamper the widespread adoption of AI technologies in the defense field, as outlined below.

Strategic and organizational barriers

Strategic muddle



Widespread AI application in the defense sphere requires a comprehensive, integrated, and coordinated strategy. Strategic unclarity leads to fragmented efforts that are not exploiting the technology’s full potential.

Bureaucratic inertia



Resistance to change within traditional defense institutions slows down AI adoption. Existing hierarchies and rigid processes (especially within procurement) hinder the flexibility required to integrate this new technology at its speed of development.

Technical and operational challenges

Data and infrastructure bottlenecks



The success of AI, regardless of the concrete application field, depends heavily on access to large, high-quality datasets and advanced computational infrastructure. Within the current defense ecosystem, AI development and deployment is significantly hindered by insufficient data collection, outdated legacy systems, strongly limited (data) interoperability, and inadequate processing capabilities.

Algorithmic reliability



While the reliability of the fundamental AI model is also central in civil applications, it is a crucial prerequisite in the defense field. Ensuring robustness and accuracy in dynamic and adversarial combat environments thus remains a central technical issue to be overcome.

User specificity



Users of AI in the defense field differ significantly from users in civilian applications. Addressing the core specific needs while retaining the key competitive edges that support the AI application in the civil domain (e.g., advanced and intuitive user experience) will be a crucial enabler for more widespread adoption.

Talent and knowledge gap

Skills shortage



The global demand for AI talent creates a highly competitive labor market, significantly hampering the ability of private but especially public defense organizations to attract and retain the skilled professionals needed for development, testing and accreditation of AI systems.

Knowledge gaps in leadership



Limited AI literacy and appreciation of its revolutionary potential in defense among some decision-makers can result in misaligned priorities and hinder effective implementation.

3.4 Risks

Apart from the vast potential, the development of AI in defense technology is associated with essential risks, requiring thoughtful evaluation to ensure they can be effectively managed and combatted.

Asymmetric threats

Non-state actors and rogue nations might leverage inexpensive AI technologies as a counter-balance to more advanced militaries and societies, in both the physical and cyber/information sphere. Examples of such asymmetrical strategies could be the deployment of a large swarm of relatively primitive AI-controlled drones that might overwhelm advanced defense systems. Apart from AI-enabled physical weapon systems, asymmetric threats could be severely amplified in the cyber and information sphere, potentially targeting critical infrastructure, communication networks, and public perception (e.g., via the mass distribution of misinformation using deepfakes) (Irregular Warfare Center, 2023).⁴⁰ Mitigating these risks requires comprehensive physical and cybersecurity capabilities, alongside civil preparedness and resilience measures.

Vulnerabilities in cyberspace

AI systems are inherently data-dependent, which creates potential critical vulnerabilities. An adversary might input false or misleading information to disturb AI-driven operations, severely affecting the ability of militaries to make informed strategic decisions, accurately assess threats, and effectively coordinate operations, thereby compromising mission success and potentially national security. While this risk applies across the board for all AI enhanced/embedded systems, it is especially relevant for advanced systems where AI systems are deeply integrated in decision-making processes.

To mitigate this risk, the development and deployment of strong cyber defense capabilities alongside the implementation of AI systems is crucial, not just to ensure the continuous operability and integrity of these technologies, but also to protect sensitive and secret data and maintain the trust and reliability necessary for their continuous and effective use.

Geopolitical turmoil

The global AI race may challenge the established order because nations often put technological supremacy ahead of traditional alliances and arms control agreements. This pursuit of dominance in AI technology can lead to increased tensions, as countries prioritize their own strategic interests over cooperative security measures. It can already be seen that there is currently no convergence on an overarching regulatory framework, for fear of losing a competitive edge in the global AI race. While global and comprehensive mitigation of this risk seems unlikely in the near-term, it serves to highlight the need to put AI at the forefront of military thinking and develop a comprehensive strategy to stay competitive in a global field. While traditional alliances may be challenged by overarching developments, these can also serve as a key multiplier and support the scaling of AI in defense technologies, providing an invaluable asset in a data-driven AI sphere. Thus, continuous close collaboration within existing alliances is crucial to maintain a strong global position.

Consequences for Germany

Particularly for Germany, navigating these target conflicts and risks demands a specific path. The focus on ethical governance and multilateralism has been an asset developed over time; however, regarding AI, this risks becoming a barrier to progress. For Germany to stay competitive, a balance must be struck between innovation and accountability in its strategy. It is crucial to build on this strong asset to become a leading force in the responsible development of AI in the defense field.



CHAPTER 4

Vision 2027+ A roadmap for Germany

Germany has reached a pivotal point in its defense strategy, where the integration of AI is no longer a matter of choice but a matter of necessity. To assume a leading position in responsible AI-driven defense, Germany needs to draw up a clear and action-oriented roadmap—one that considers its challenges and plays to its strengths. This vision, for 2027 and beyond, is anchored on four interoperating priorities, namely: AI sovereignty, NATO and EU interoperability, innovation ecosystems, and leadership in ethical AI governance.

The first aspiration for Germany should be to develop a certain degree of AI sovereignty, entailing the national development of critical technologies. As such, sovereignty means reducing dependence on foreign providers for systems forming the backbone of national security. As a priority, Germany needs to develop a strong overall ecosystem in support of autonomous systems, AI-driven logistics, and advanced cyber defense platforms. This implies intensive investment in R&D and effective collaboration between government, private enterprise, and academia. For example, in 2023, the EDF has spent 367 mn. EUR on European collaborative defense R&D projects (EDA, 2024). Also, France has introduced the “l’agence ministérielle pour l’IA de défense”, a ministerial agency for defense AI, with a budget of 300 mn. EUR (Ministère des Armées, 2024). Historically, Germany has focused on industrial precision and technological innovation, which can be leveraged as a springboard for developing AI capabilities attuned to its very specific defense needs.

No less important is the need to maintain interoperability with NATO and EU allies. Being a central contributor to these alliances, Germany should make sure its AI systems integrate smoothly into joint operations. For such interoperability, it will be necessary to follow shared technical standards, including data-sharing protocols and systems compatibility. Germany’s defense planners must also prioritize participation in multinational AI exercises, which provide opportunities to test systems, refine strategies, and strengthen trust among allies. Leading efforts to establish NATO-wide AI frameworks could position Germany as a key innovator and influencer within the alliance.



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Innovation ecosystems are another critical aspect of this roadmap. Germany has a world-class research infrastructure and a vibrant private technology sector. The gap between civilian innovation and defense applications is tremendous. This divide can only be filled if the government enables and encourages collaboration amongst startups, established firms in this industry, and the military entities. Regional innovation hubs can be developed where companies and research institutions will focus exclusively on AI projects with a defense orientation. These hubs should be underpinned by easy, seamless procurement processes to enable the rapid development and fielding of technologies that meet military requirements.

Germany also needs to lead on ethical AI governance. The country's reputation for human rights and international law makes it well-positioned to determine the gold standard for responsible military AI globally. Embedding ethics into all levels of AI development, from data collection to deployment, Germany can position itself as compliant with international humanitarian law. Initiatives such as a national AI Ethics Board for Military Technologies and transparent audit processes are very much integral in this regard. This ethical leadership can also act as a counterbalance to the heavy-handedness of countries such as China and Russia, where nations put speed and efficiency above accountability.

This vision should be part of both near-term challenges and long-term innovation to 2027. Quantum AI, swarm robotics, and space-based AI systems will be the emerging technologies beyond 2030 that will shape the global defense landscape. It would, therefore, be prudent for Germany to position itself in a leadership role in these areas now, to guarantee relevance in the coming decades.

Milestones towards 2027

Progress in realizing these aspirations will be a phase-by-phase process. This is where, between now and 2027, Germany must focus on creating the enabling environment for AI integration, testing pilot projects, and scaling successful initiatives into full operational capabilities (see *Exhibit 3*).

EXHIBIT 3 Roadmap for Germany



Source: Strategy& analysis

Overarching: Address cultural and ethical challenges

Public acceptance of AI in defense necessitates transparency about its benefits, such as reducing casualties and improving mission precision, but also addressing the inherent target conflicts and risks. Clear and interdisciplinary ethical oversight will be critical to ensure the legitimacy of Germany's AI defense strategy, ensuring that AI systems adhere to legal and moral standards.

1. Formulate an integrated AI strategy in defense

Germany must centralize decision-making to eliminate fragmentation. Thus, Germany should establish a National AI Defense Council under the Ministry of Defense that will coordinate AI activities, prioritize resources, and ensure accountability. The council should develop a national AI strategy aligning defense objectives with investment in key areas like autonomous systems and predictive maintenance. Public-private partnerships and collaboration with EU partners are essential in this regard.

2. Invest in high-performance infrastructure

Modern AI systems demand robust infrastructure. Thus, Germany needs to focus on building secure data centers and edge computing that can meet the demanding requirements of the military. All such investment should be complemented by enhanced cybersecurity measures to safeguard AI systems against adversarial attacks. The implementation of dual-use infrastructure, as outlined in primary NATO initiatives, will be a key enabler here.

3. Build a sustainable talent pool

Addressing the talent shortage in this particular field is a critical enabler to reach the set objectives. Germany should create Defense AI fellowships to fund research and innovation in universities, establish a Defense AI Academy, and partner with civilian AI leaders like Siemens and SAP for joint training programs. Competitive incentives will attract skilled professionals to defense roles.

4. Reform procurement processes

Germany's procurement system needs modernization to match AI's rapid development cycles. Agile procurement models, sandbox environments for testing, and a Defense Innovation Fund will expedite technology adoption. Initial AI deployments should focus on low-risk areas like logistics, gradually expanding to critical applications.

5. Enhance NATO and EU cooperation

AI integration requires international collaboration. Germany should advocate for standardized AI protocols, co-invest in EU research hubs, and lead joint military exercises. Establishing a European Defense AI Consortium will further bolster shared innovation.

Implementation framework

By following this roadmap, Germany can establish itself as a leader in responsible AI in defense development, aligning operational effectiveness with ethical standards.



Phase 1

Planning and coordination

Establish the AI Task Force, finalize the national strategy, and begin talent and infrastructure development.

Phase 2

Initial deployment and testing

Pilot AI-driven logistics and surveillance, validate interoperability in NATO exercises, and graduate the first AI Academy cohort.

Phase 3

Full-scale implementation

Deploy battlefield-ready AI systems, integrate with NATO and EU frameworks, and refine strategies through annual evaluations.



CONCLUSION

Artificial intelligence is not just a technological development, but rather a paradigmatic structural change in how countries think about defense, strategy, and security in the 21st century.

By 2030, the integration of AI technologies across predictive decision-making, collaborative autonomous systems, and dynamic resource management is anticipated to redefine military operations, offering unprecedented precision, agility, and resilience.

However, the transformative power of AI also brings heightened ethical, legal, and strategic complexities that demand vigilant oversight and deliberate governance: (uncoordinated) proliferation of autonomous systems, ethical dilemmas of machine-led warfare, and vulnerabilities arising from heavy reliance on complex, interconnected systems.

To harness AI's full potential while mitigating the associated risks, defense organizations must prioritize the development of robust ethical frameworks, transparent accountability mechanisms, and international collaboration. These measures will not only ensure the responsible use of AI, but also maintain trust and legitimacy within the global security ecosystem. The journey towards AI-driven defense capabilities will require a careful balance between innovation and regulation, with a shared commitment to advancing technology in a manner that upholds the principles of democratic values and human rights.

Germany stands at a pivotal crossroads in integrating AI into its defense strategy. To remain a relevant military power and a key contributor to NATO and the EU, Germany must act decisively to overcome institutional fragmentation, cultural resistance, and underinvestment in talent and infrastructure. Leveraging its world-class research institutions, industrial expertise, and international partnerships, Germany has the potential to build an AI defense ecosystem grounded in ethical governance and innovation. Immediate priorities include securing infrastructure, recruiting AI specialists, and aligning national (and AI) defense strategies with broader international objectives.

The global AI race is heating up, and the battlefield of tomorrow is being molded today. Germany needs to make sure it is not lagging behind, but stands at the forefront of innovative, ethical, AI-driven defense. This is more than a strategic necessity; it's a national must. And the time to act is now.



The journey towards AI-driven defense capabilities will require a careful balance between innovation and regulation, with a shared commitment to advancing technology in a manner that upholds the principles of democratic values and human rights.”

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